MIL-S-8602(SHIPS)
21 [illegible] 1966
SUPERSEDING
MIL-S-860A(SHIPS)
28 March 1955
(See 6.2)

MILITARY SPECIFICATION

STEEL FORGINGS FOR STEAM TURBINE ROTORS

1. SCOPE

1.1 Scope. - This specification covers steel turbine rotor forgings for use in steam atmospheres which do not exceed the rated operating temperatures specified in 1.2.

1.2 Classification. - Steel forgings shall be of the following grades, as specified (see 6.1):

	Maximum operating temperature
Grade A - Carbon-Steel	650°F.
Grade B - Nickel-molybdenum-vanadium	750°F.
Grade C - Nickel-chromium-molybdenum-vanadium	750°F.
Grade D - Nickel-chromium-molybdenum-vanadium	750°F.
Grade E - Nickel-chromium-molybdenum-vanadium	750°F.
Grade F - Chromium-molybdenum-vanadium	1050°F.
Grade G - 12 percent chromium	750°F.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

STANDARDS

FEDERAL
FED-STD-151 - Metals; Test Methods.

MILITARY
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-163 - Steel Mill Products, Preparation for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC FORG

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2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM A443 - Notch Toughness of Turbine and Generator Steel Forgings.
ASTM A472 - Heat Stability Test for Steam Turbine Shafts and Rotor Forgings.
ASTM E208 - Drop Weight Test For Nil Ductility Temperature Of Steels.

(Application for copies should be addressed to the American Society For Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical society and technical association specifications and Standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material. - The ingots used for the forgings shall be made from steel, produced in the electric furnace, which has been vacuum treated prior to teeming.

3.1.1 Discard. - Sufficient discard shall be taken from each ingot to insure freedom from piping and segregation in excess of that permitted by the check analysis variation specified in 3.5.1.

3.2 Forging process. - To insure proper working of the metal, the reduction in area from the ingot to the main body of the forging shall be at least 3 to 1, however, in the case of forgings over 1200 square inches in cross section a reduction of 2-1/2 to 1 will suffice.

3.2.1 Dimensions. - All forgings shall conform to the dimensions specified in the contract or order (see 6.1).

3.3 Heat treatment. - All forgings shall be given such uniform heat treatment to produce material conforming to this specification. Heat treatment shall consist of quenching, and tempering or normalizing and tempering. No straightening by any means is permissible after heat treatment for mechanical properties. Minimum tempering temperatures shall be as follows:

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Grade	Temperature
A	1100°F.
B	1100°F.
C	1100°F.
D	1100°F.
E	1100°F.
F	1100°F.
G	1200°F.

3.3.2 After heat treatment for mechanical properties and prior to heat stability testing, the forgings shall be rough machined completely and then stress relieved at a temperature 50°F. to 100°F. below the final tempering temperature. The forging shall be held at the stress relief temperature for a period of 2 hours per inch of maximum radial thickness.

3.3.3 The contractor shall maintain records of all heat-treatments and shall make them available upon request (see 6.1).

3.4 Boring. - When boring is specified (see 6.1), the forging shall be bored prior to heat treatment for mechanical properties.

3.5 Chemical composition. - The chemical composition of the forging shall conform to table I.

Table I Chemical composition (ladle analysis) 1/

Element	Grade A	Grade B	Grade C	Grade D	Grade E	Grade F	Grade G
Carbon	.45	.25	.28	.28	.28	.25/.35	.15
Manganese	.90	.20/.60	.20/.60	.20/.60	.20/60	1.00	1.00
Phosphorus	.025	.015	.015	.015	.015	.015	.015
Sulphur	.025	.018	.018	.018	.018	.018	.018
Silicon	.15/.35	.15/.30	.15/.30	.15/.30	.15/.30	.15/.35	.50
Nickel	---	2.50 min	2.50 min	3.25/4.00	3.25/4.00	.75	.50
Chromium	---	.75	.75	1.25/2.00	1.25/2.00	.90/1.50	11.5/13.0
Molybdenum	---	.25 min	.25 min	.30/.60	.30/.60	1.00/1.50	.50
Vanadium	.03/.12 2/	.03 min	.03 min	.05/.15	.05/.15	.20/.30	---

1/ Weight percents are maxima unless a range is shown or otherwise noted.
2/ Vanadium is not mandatory. When vanadium is used it shall conform to the specified range.

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3.5.1 Check analysis variations. - Check analysis variations of the elements shall not be over the upper and under the lower limits by more than the percentages specified in table II.

Table II - Permissible check analysis variations in weight percent.

Elements	Cross-sectional area, square inches			
	200 or less	Over 200 - 400	Over 400 - 800	Over 800
Carbon	.03	.04	.05	.06
Manganese	.04	.05	.06	.07
Phosphorus	None	None	None	None
Sulphur	None	None	None	None
Silicon	.02	.03	.04	.04
Nickel	.07	.07	.07	.07
Molybdenum	.04	.04	.05	.06
Chromium	.04	.04	.05	.05
Vanadium	.01	.01	.01	.02

3.6 Mechanical properties. - The forgings shall meet the mechanical property requirements specified in table III.

Table III - Mechanical properties.

	Grade A	Grade B	Grade C	Grade D	Grade E	Grade F	Grade G
Tensile strength, psi, min[1]	75,000	80,000	90,000	105,000	120,000	115,000	95,000
Yield strength, psi, min. (0.02 percent offset)[2]	40,000	55,000	70,000	85,000	95,000	85,000	70,000
Percent elongation, 2 in.[2]							
Longitudinal, min.	22	22	20	18	18	17	18
Radial, min.	20	20	17	17	17	14	16
Percent reduction of area[2]							
Longitudinal, min.	40	50	48	52	52	43	45
Radial, min.	30	50	45	56	50	38	50
Impact, Charpy V-notch, min[3] (ft.-lbs. at 70°F.)	10	28	25	45	40	6	15
Transition temperature, °F. max.[4]	175	100	110	40	50	250	140

[1] All radial and longitudinal tensile tests shall meet these requirements.
[2] When more than one test is required, as in forgings weighing more than 5000 pounds, all tests shall meet these requirements.
[3] Average of the tests required.
[4] Based on 50 percent fibrous fracture appearance of Charpy V-notch tests.

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3.7 Nil ductility temperature. - The nil ductility temperature for each forging shall be determined for information purposes. The nil ductility temperature as specified in ASTM E-208 shall be forwarded to the Commander, Naval Ship Engineering Center, Code 6641, Washington, D.C. 20360.

3.8 Nondestructive testing. -

3.8.1 Ultrasonic examination. - Each forging shall be ultrasonically examined to determine the acceptance criteria specified in Appendix A.

3.8.2 Magnetic particle examination. - Each forging shall be magnetic particle tested for conformance to the acceptance criteria of Appendix A.

3.8.3 Boroscopic examination. - Each bored forging shall be examined by use of the boroscope as specified in Appendix A.

3.8.4 Stability. - Unless otherwise specified (see 6.1), each forging shall be subjected to a heat indication test to determine its stability or freedom from tendency to distort during high temperature operating conditions. Maximum excursion from the final hot reading to the final cold reading shall not exceed .001 inch at any point.

3.9 Marking. - Each forging shall be legibly marked with the forging and heat number, specification number, grade, contract or order numbers, and with such other information as may be required (see 6.1).

3.10 Workmanship. - The forging shall be free from defects such as seams, pipes, scale, fins, porosity, hard spots, excessive nonmetallic inclusions and from thermal cracks and flakes.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling. -

4.2.1 Chemical or spectrographic analysis. -

4.2.1.1 Ladle analysis. - An analysis of each heat shall be made by the manufacturer. The test specimen shall be taken during the pouring of the heat.

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4.2.1.2 Check analysis. - One check analysis shall be made on one forging from each heat. The sample may be taken from a broken tensile specimen.

4.2.2 Sampling for mechanical tests. -

4.2.2.1 Prolongations. - Unless other provisions are made for test specimens with approval of the Command or agency or its designated representative, the forgings shall be furnished with suitable prolongations for the required test samples. The prolongations shall be approximately the same diameter as the main body of the forging and shall be at least 1/2 of the main body radius in length.

4.2.2.2 Location of specimens. -

4.2.2.2.1 Radial and longitudinal tension. - Radial and longitudinal tension specimens shall be selected from the prolongation as near as possible to the prolongation mid-radius and a distance from the end equal to 1/4 of the prolongation radius. Radial tension specimen shall be taken from the opposite end for forgings weighing over 5000 pounds.

4.2.2.2.2 Impact and transition temperature. - The charpy V-notch specimens shall be selected adjacent to the radial tension specimen. The length of the specimen shall be oriented radially. Drop weight specimens shall be selected as specified in ASTM E-208, with their geometric centers located in the prolongation radius and at a distance of 1/4 the prolongation radius from the end.

4.2.2.3 Number of specimens. - For forgings weighing less than 5000 pounds, one longitudinal tension specimen, one radial tension specimen, at least four charpy V-notch test specimens, and at least four ASTM E-208 drop weight test specimens shall be taken from a prolongation as specified in 4.2.2.1.

4.2.2.3.1 For forgings weighing more than 5000 pounds, one longitudinal tension specimen, one radial tension specimen, at least four charpy V-notch specimens and at least four ASTM E-208 drop weight test specimens shall be taken from a prolongation as specified in 4.2.2.1. In addition, one radial tension specimen and at least four charpy V-notch specimens shall be taken from the opposite end by trepanning if desired.

4.2.2.4 Types of specimens. - The tension and radial test specimens shall conform to type H of method 211.1 of FED-STD-151 and the charpy V-notch specimen shall conform to figure 3 of method 221.1 of FED-STD-151. The drop weight specimen shall conform to type P-2 of ASTM E-208.

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4.3 Examination. -

4.3.1 Dimensional examination. - Each forging shall be subjected to dimensional examination. Forgings with nonconforming dimensions shall be referred to the Command or agency for resolution and acceptance.

4.3.2 Magnetic particle inspection. - The forgings shall be examined by the magnetic particle inspection in accordance with Appendix A to meet the acceptance criteria requirements therein. This inspection shall be performed after all machining is accomplished.

4.3.3 Ultrasonic examination. - The forgings shall be examined as specified in Appendix A to meet the acceptance criteria requirements therein.

4.4 Tests. -

4.4.1 Chemical or spectrographic analysis. - The samples required by 4.2.1.1 shall be analyzed in accordance with method 111.1 or 112.1 of FED-STD-151 to determine whether the composition of the steel is in accordance with table I. Check analysis, as required by 4.2.1.2, shall also meet the requirements of table I with the variations of table II.

4.4.2 Tension, radial, and impact test. - The number and type of specimens specified in 4.2.2.3 and 4.2.2.4 shall be tested as specified in methods 211.1 and 221.1 of FED-STD-151 to determine conformance to the requirements of table III (see 3.6).

4.4.2.1 Transition temperature impact test. - The transition temperature shall be determined by the method specified in ASTM A-443 and shall meet the requirements of 3.6.

4.4.3 Nil ductility temperature determination. - For determination of the nil ductility transformation temperature, by the drop weight test method, ASTM E-208 shall be employed.

4.4.4 Stability or heat indication test. - Each forging shall be subjected to a heat indication test and meet the requirements of 3.8.4. The testing shall be conducted after heat treatment, machining, and stress relieving and shall be in accordance with ASTM A-472. When the Command or agency considers an alloy and geometry to be insensitive to dimensional change at operating temperatures, the Command or agency reserves the right to waive this test. In such cases the Command or agency has the responsibility for the forging stability.

4.5 Rejection. - Forgings failing to conform to the requirements of this specification shall be rejected.

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4.6 Reheat treatment. - When any forging fails to conform to 3.6, 3.7, or 3.8 it may be reheat treated and subsequently retested as applicable.

5. PREPARATION FOR DELIVERY

5.1 Packing. -

5.1.1 For domestic shipment, immediate use. - The forgings shall be prepared for shipment in a manner to insure carrier acceptance at the lowest rate and safe delivery at destination. Containers shall conform to the Uniform Freight Classification Rules or regulations of other carriers as applicable to the mode of transportation.

5.1.2 For domestic shipment and storage and overseas shipment. - The forgings shall be prepared for shipment in accordance with MIL-STD-163.

5.2 Marking. - In addition to any special markings required by the contract or order, marking for shipment shall be in accordance with MIL-STD-129 (see 6.1).

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) The grade of forging required and the rated operating temperature (see 1.2).
(c) Production drawings including dimensions and tolerances (see 3.2.1).
(d) When heat treating data is required (see 3.3.3).
(e) If forgings are to be other than bored (see 3.4).
(f) When determination of stability is not required (see 3.8.4).
(g) Additional identification marking if required (see 3.9).
(h) Preservation, packaging, and marking requirements, if other than those specified (see 5.1 and 5.2).

6.2 CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

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APPENDIX A

10. SCOPE

10.1 Scope. - This appendix constitutes the minimal requirements for nondestructive testing vacuum treated steel forgings for steam turbine rotors. It does not intend to restrict the supplemental use of other tests by a turbine manufacturer.

20. APPLICABLE DOCUMENTS

20.1 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM E-127 - Recommended Practice For Fabricating and Checking Aluminum and Aluminum Alloy Ultrasonic Standard Reference Blocks.

(Application for copies should be addressed to the American Society For Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

30. MAGNETIC PARTICLE TESTING REQUIREMENTS

30.1 Each forging shall be magnetic particle tested over 100 percen of the finished forging surface in accordance with the requirements specified herein.

30.2 Personnel requirement. - Personnel performing and interpreting magnetic particle testing shall have demonstrated an ability to perform an operational type test using the test technique which is to be used in production. In addition the operator shall be familiar with the magnetic particle test requirement and acceptance-rejection standards of this specification.

30.3 Equipment and material requirements. -

30.3.1 Magnetizing apparatus. - The magnetizing apparatus shall be capable of inducing in the forging under test, a magnetic flux of suitable intensity to delineate tight discontinuities (cracks and nonmetallic inclusions) 1/16 inch and over in length that lie up to 1/16 inch below the testing surface.

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30.3.2 Magnetic particles. - The magnetic particles used for test shall be finely divided ferro-magnetic material of high permeability and low retentivity. The particles shall be free from foreign material which may be deleterious to their function.

30.3.3 Vehicle. - The liquid used as a vehicle for magnetic particles for test by the wet method shall comply with one of the following:

(a) P-S-661 - Solvent dry cleaning.
(b) VV-K-211 - Kerosene.
(c) Tap water with suitable rust inhibitor, wetting and antifoaming agents. Composition shall be approximately 0.3 percent antifoam agent, 4 percent rust inhibitor, 13 percent wetting agent, balance tap water.

30.3.4 Magnetic particle test procedure. - A documented test procedure shall be employed. The written procedure shall be made available to the Command or agency concerned. At minimum it shall contain the following:

(a) Type or description of the magnetizing apparatus.
(b) Type or description of magnetic particles.
(c) Type of liquid vehicle when wet method is employed.
(d) Surface preparation - The procedure shall contain the surface finish of the tested product. The surface shall be clean and free from foreign material.
(e) Direction of magnetization - To insure detection of discontinuities having axes in any direction, the procedure shall describe a method of testing employing magnetic fields in two directions the second of which shall be at right angles to the first.
(f) Magnetizing current and conditions of magnetization shall be a part of the procedure. These shall be adjusted to assure the delineation of tight discontinuities (cracks and nonmetallic inclusions) 1/16 inch and over in length that lie up to 1/16 inch below the testing surface.
(g) Demagnetization - A method for demagnetization shall be a part of the procedure. Demagnetization shall be employed prior to testing if the forging contains remnant fields from a previous operation, and on the finished forging when the remnant field interferes with removal of particles or a subsequent use of the forging.
(h) Cleaning - The procedure shall specify the method of cleaning subsequent to test.

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30.3.5 Acceptance-rejection standards. - All magnetic particle test indications over 1/8 inch long shall be determined to be free of cracks. Random indications up to 1/2 inch long due to nonmetallic inclusions are acceptable. Indications over 1/2 inch long due to nonmetallic inclusions and clusters (over 6 indications in 4 square inches) of nonmetallic inclusions up to 1/2 inch long shall be reported to the Command or agency for resolution of their acceptability.

30.3.6 Report of magnetic particle test. - A report of the magnetic particle test shall be forwarded to the Command or agency concerned.

40. BOROSCOPIC TEST REQUIREMENT

40.1 Each bored forging shall be examined by use of the boroscope.

40.2 Personnel performing this test shall have demonstrated their responsibility and shall be capable of discerning small defects in the bore of quality forgings.

40.3 All defects delineated by this inspection shall be reported to the Command or agency for resolution. Acceptability of forgings with bore defects rests with the Command or agency. A defect is defined for these as any discontinuity in the metal surface which is visible by the employment of the boroscope.

40.4 A report of the boroscopic examination shall be forwarded to the Command or agency concerned.

50. ULTRASONIC TESTING REQUIREMENTS

50.1 The requirements set forth herein are the minimum acceptable for a satisfactory test of internal quality of forgings produced to this specification. These requirements shall in no way restrict the use of supplemental ultrasonic tests nor restrict the use of new improved ultrasonic test equipment and methods when they become available.

50.2 Personnel requirements. - Personnel performing and interpreting ultrasonic tests shall have demonstrated an ability to perform an operational type test using the test technique to be used in production. Operators shall have demonstrated their ability to calibrate the instrument for test purposes and shall have a general knowledge of the effects of surface finish, grain structure, transducer size and type, and testing frequency on the results of test. Operators shall have a working knowledge of the ultrasonic test requirements and acceptance-rejection standards of this specification.

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50.3 Ultrasonic test equipment. - The equipment and material employed in the ultrasonic test of forgings shall be as follows:

(a) Electronic apparatus - The ultrasonic equipment shall be capable of producing, receiving, amplifying and displaying electrical pulses at 1 to 50 megacycles (mc). The instrument shall be of the pulse-echo type with a method of data presentation on a cathode ray tube utilizing a horizontal base line which indicates elapsed time when reading from left to right and vertical deflection in the base line indicates signal amplitude ("A" scan presentation). Further, the instrument shall have circuitry to provide a continuously increasing amplification with respect to time which is compensating for the reduction in sensitivity with depth as a result of sound attenuation (automatic gain circuitry).

(b) Transducer - The transducer shall be a quartz crystal capable of carrying out the operational test at 2.25 mc. The transducer active element shall be nominally 1 square inch (1-1/8 inch diameter round crystal).

(c) Liquid couplant - The liquid couplant, causing acoustic coupling between the transducer and forging, shall not be injurious to the forging and shall have a viscosity no greater than that of SAE 30 motor oil.

50.4 Calibration of ultrasonic test equipment. - The instrument shall be permitted to warm up before calibration is attempted. Using a frequency of 2.25 mc. the equipment shall be calibrated with block number 3-0600 of ASTM E-127 calibration standard each time it is used, and shall be re-calibrated whenever electric power to the instrument has been interrupted of when a different diameter of forging is to be inspected. The signal amplitude from a 3/16 inch flat bottom hole shall coincide with the following when calibrating for testing the corresponding forging diameters:

Forging diameter (inches)	Indication amplitude sweep to peak percent of screen height
15 - 20 excl.	57
20 - 25 excl.	52
25 - 30 excl.	48
30 - 35 excl.	42
35 - 40 excl.	39
40 - 45 excl.	36
45 - 50 excl.	33
50 - 55 excl.	31
55 - 60 excl.	29

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50.5 Ultrasonic test procedure. - At minimum, the following shall be employed in an ultrasonic test procedure. The procedure shall be documented and shall be made available to the Command or agency concerned upon request:

(a) The surface condition of the forging shall be 250 root mean square (rms), or better prior to testing. Further the surface shall be free from foreign material which would interfere with a normal ultrasonic test.

(b) Forgings shall be tested either while stationary or while being rotated to a peripheral speed of not more than 200 inches per minute. The forging shall be tested such that sound is passed radially through the forging and 100 percent of forging volume is covered.

(c) For each diameter of the forging under test the instrument as calibrated (see 50.4) shall be employed. The transducer shall be brought into contact with the forging surface and the sweep length shall be adjusted such that the resultant signal from the first back reflection is approximately 3/4 of the distance from left to right across the cathode ray tube.

(d) The ultrasonic test shall then be performed over 100 percent of the forgings circumferential area. Each transducer path shall overlap the previous transducer path by 50 percent of the transducer diameter.

(e) Each indication shall be classified as "stationary" or "traveling". Stationary indications shall be defined as any indication which, when exhibited on the cathode ray tube, remains stationary on the sweep line with minor transducer movement. A traveling indication shall be defined as any indication which, when exhibited on the cathode ray tube, travels either to the left or right on the sweep line with minor transducer movement.

(f) All indications shall be noted as stationary or traveling and recorded in terms of percentage of screen height. All areas exhibiting loss of back reflection in excess of 20 percent of the normal back reflection shall be examined at other angulations (excluding shear wave) to determine whether there are indications present.

(g) An ultrasonic test record shall include a comprehensive description in graphic form of the involved forging showing locations of all indications in excess of the response from the calibration standard. The record shall note the type of indication (traveling or stationary) and all areas showing 20 percent or greater loss of normal back reflection.

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50.6 Ultrasonic acceptance-rejection standards. - The acceptability of forgings produced to this specification shall be judged from figures A1 and A2. Figure A1 is a plot of the acceptability line for the combination of stationary and traveling indications. The total number of recordable indications versus magnitude shall be plotted on figure A1. If the test result plot lies within the area designated acceptable, a plot of the number of recordable traveling indications, versus magnitude shall be made on figure A2. If this plot lies within the area designated acceptable, the forging is acceptable. Where part of either plot in figure A1 and A2 is outside the area of acceptability, the test data shall be referred to the Command or agency. Acceptability of such data is the responsibility of the Command or agency concerned. Where the entire plot lies outside the area of acceptability of either figures A1 or A2 the forging shall be rejected.

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FIG. A-1 TOTAL INDICATIONS VS MAGNITUDE

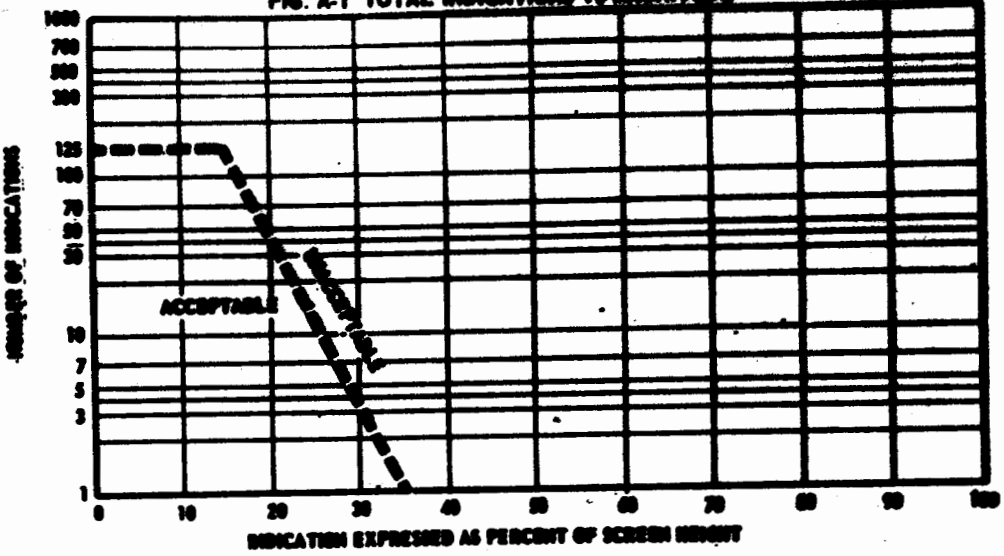
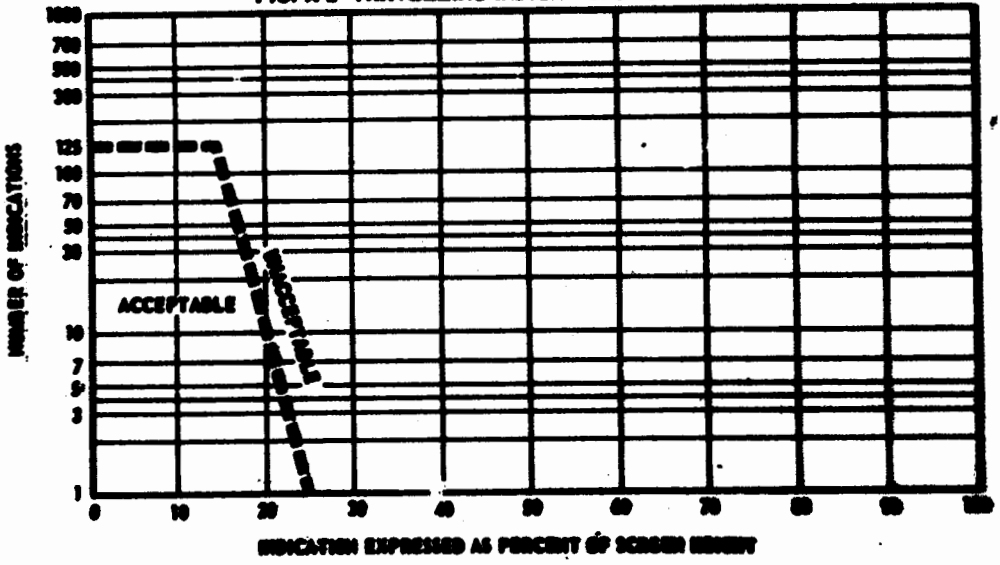


FIG. A-2 TRAVELLING INDICATIONS VS MAGNITUDE



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